DIC05
Miller Capacitance

Specification-based laboratory report

17 October 2026

English edition
HSPICE S-2021.09 | Model library supplied separately

1 Objectives

Follow Lab 5 (PDF pages 21–23) to observe Miller coupling at the second stage of a three-inverter chain and quantify rail excursions and loading.

2 Specification mapping

Handout requirement (PDF page)	Implementation	Verification and scope
Step 1: three inverters and stage-2 glitches (21–23)	01_miller, case 0	Simulated with native parasitics
Step 2: add and vary C_c (22–23)	Cases 1–3: 2/5/10 fF	Includes the specified 5 fF
Question: increase output load (23)	Cases 4–5: add 10/40 fF	Compare at fixed 5 fF coupling

3 Circuit and design derivation

All three inverters use $W_n/W_p/L = 0.6/1.8/0.25\,\mu$m. C_c spans the second-stage input and output. Stage 3 remains a gate load, while additional C_L connects to ground. The qualitative relation $\Delta V_o \approx C_c \Delta V_i/(C_c + C_{\text{out}})$ explains charge division; nonlinear MOS simulation supplies the actual values. Scattered source cases are replaced by comparable parameter cases with explicit peak and edge measurements.

4 Simulation setup

This teaching rewrite uses a separately supplied model at 1.0 V and 25 °C. The specified topology, loads and comparative dimensions are retained. The handout's 2.5 V process results are not acceptance values for this model. The chosen 1 V is a simulation condition, not a verified device rating. MOS bulk terminals are explicitly connected to the corresponding supply rails.

HSPICE S-2021.09 uses RUNLVL=6, retaining waveforms, printed data and native measurements. Figures are redrawn from actual data; CSV files are in each Lab's results directory. Successful solution does not establish process reliability. Warnings from encrypted model blocks remain suppressed; the existing diagnostic did not expose actionable further details, so the warnings are not declared harmless and complete model-limit compliance is not claimed.

Input edges are 0.1 ns, with a 1 ns initial delay, 2 ns high interval and 5 ns period. Transient duration is 10 ns with 1 ps printing. Peaks are measured over 0.8–5 ns; rise/fall use 10–90%/90–10% crossings.

5 Results and specification checks

All six cases concluded normally without errors or failed measurements. C_L is additional to the stage-3 gate load.

C_c (fF)	C_L (fF)	Over (mV)	Under (mV)	t_r (ps)	t_f (ps)
0	0	123.09	83.47	133.70	76.70
2	0	154.03	130.47	155.08	99.82
5	0	188.80	182.72	189.49	133.95
10	0	227.81	237.61	250.08	190.65
5	10	143.98	142.61	279.78	178.15
5	40	80.57	80.85	558.95	318.36

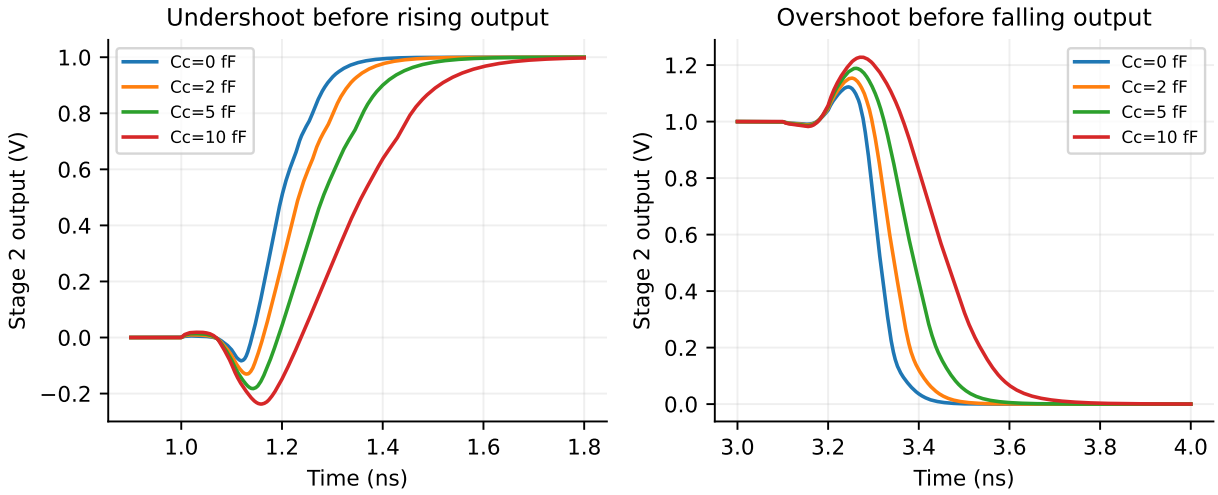


Figure 1: Transitions and rail excursions at stage 2; redrawn from this run's HSPICE data.

6 Discussion

Increasing C_c from 0 to 10 fF raises overshoot from 123.09 to 227.81 mV and undershoot from 83.47 to 237.61 mV. At 5 fF coupling, adding 40 fF load reduces the peaks to about 80.6/80.9 mV while increasing rise/fall times to 559/318 ps. Zero added coupling does not remove intrinsic Miller capacitance. Rail excursions are simulation results; they do not establish hardware reliability or immunity of every receiver.

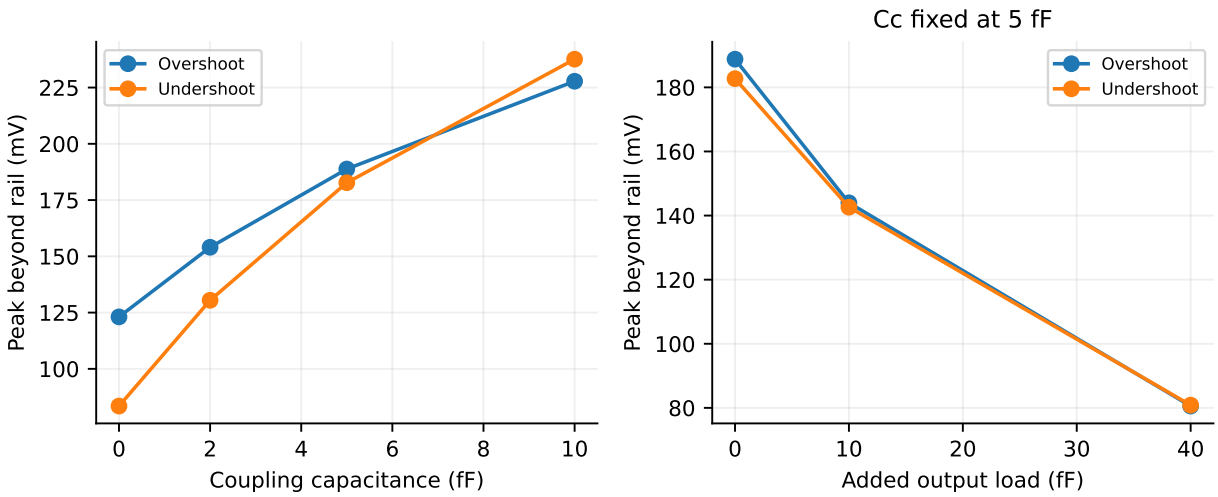


Figure 2: Effect of coupling capacitance and additional output load; redrawn from this run's HSPICE data.

7 Conclusion

Both experimental steps and the loading question are covered by actual data. Results support capacitive coupling and charge-division reasoning under the stated model and stimulus.